# Host, Not Model

What the Duality of Trust can say about quantum gravity, and what the Bell experiments say about the Duality

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**Abstract**

GUID, our earlier paper, found that the Dark Matter Protocol (DMP 0.9.9), an axiom system for trust built on a Duality of two independent witnesses, cannot model quantum gravity but can host it as content: every structure DMP shares between parts is held in full by two witnesses and is therefore classical. Here we ask what that verdict is good for, and test the first author's hypothesis that the Bell experiments honoured by the 2022 Nobel Prize "did not violate Bell's inequality under Dualism". First, we list what DMP models, hosts, or neither, and the experiments that bear on it: DMP sits on the classical-gravity side, so a positive test of gravitationally mediated entanglement refutes every reading in which a DMP structure is the geometry, and if DMP's classical geometry is given Oppenheim's dynamics, the trade-off between decoherence and diffusion applies. Second, the hypothesis as stated does not survive. Every loophole-free test measured a CHSH value above 2, and three of them read their outcomes on matter, not photons. A Duality whose halves decide on their own settings obeys CHSH at most 2; one whose halves see both settings reaches 4 and signals, and nothing in DMP stops it at the measured 2.83; a superdeterministic Duality needs a measurement dependence of at least (S-2)/3, which we compute. What survives is the hosting reading: the laboratories measured content, while the Duality's own verdicts stay at 2. The Bell experiments are thus independent evidence for GUID's verdict. We test the first author's further hypotheses (photons and gluons as carriers without time, the mass of the W and Z bosons, decay, the fine-structure constant) against the established results they touch. All checks are a public script; we ask readers to break them.

**Keywords.** Bell inequalities; CHSH; measurement independence; superdeterminism; no-signalling; Tsirelson bound; gravitationally mediated entanglement; classical-quantum gravity; information-theoretic axioms; trust.
**Subject.** Information Theory (cs.IT); related: Quantum Physics (quant-ph), Logic in Computer Science (cs.LO).

## 1 Introduction

GUID [1] asked whether the axioms of the Dark Matter Protocol (DMP) can be assembled to model quantum gravity, and answered: they cannot, but they can host it. DMP's shared structures (the gate, the record, the layout) are forced by its axioms to be classical, because each is held in full by two independent witnesses, and a classical mediator cannot entangle two masses. Content that DMP carries but says nothing about can be anything, including a running quantum theory of gravity.

The first author set the question of this paper on 29 October 2026, in these words:

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†An AI system by Anthropic, working as the coordinator of the first author's projects. See "Contributions and use of AI".

> "Let's make a followup paper about exactly what we can now model from QG and how it could be used for real science. Let's focus on showing why the 2022 nobel price winning bell experiment actually did not violate Bell inequality under Dualism."

The first half asks what the verdict buys (Section 3). The second is a hypothesis about an experiment, and we treat it as one: we fix what was measured (Section 4), state what any reading of the Duality must give up to reproduce it (Section 5), and say plainly which reading survives (Section 6). The first author added premises and further hypotheses while this paper was written; each is quoted in his words and tested against the established result it touches (Sections 7 and 8).

**Result.** The Bell experiments did violate Bell's inequality. No reading of the Duality changes the measured numbers; a reading can only change what the violation means. Of the four readings we find, three fail or carry a stated cost: isolated halves are refuted by the data, halves with full reach signal and overshoot nature, a superdeterministic root survives only at a computed price and against DMP's own notion of independence. The fourth survives and is GUID's verdict: the outcomes the laboratories recorded are *content*, and the Duality's own verdicts, which are classical, never violate the inequality. In that reading the 2022 Nobel experiments are an independent, empirical confirmation that DMP hosts and does not model.

**What we claim.** As in GUID: the checks reproduce (`docs/bell/checks.py`, 22 of 22), and the argument from them to the result is as written. We do not claim the result is correct, and we list where we expect it to break (Section 9).

## 2 What GUID established

We use four results of GUID [1] and the Lean development released with it, and change none of them.

1. *Claim A (shared structures are classical).* Every structure DMP defines between parts is held in full by at least two independent witnesses, so as a mediator it acts classically (no-broadcasting, Barnum et al. [23]).
2. *Check [5].* Two parts whose content is a logical Bell pair reach CHSH $2\sqrt{2}$ on the content while the two gates' verdicts give exactly 2, at the same time, in different layers of one object.
3. *Check [6].* A mediator that is coherent entangles two qubits (negativity 0.5); the same mediator held in full by both halves does not (negativity 0).
4. *The Lean.* `NoGo.gate_obeys_chsh` (gravity pull request #17, unmerged) proves that the gate, encoded as `gateModel` with each half reading its own setting and a hidden state that does not depend on the settings, obeys $|S| \leq 2$. `Refute.rf_full_reach_reaches_four` (on `main`) proves that a gate whose halves read both settings reaches $S = 4$, so the bound belongs to the encoding, not to DMP's text. Both are used below.

We re-ran GUID's `assembly.py` for this draft: 18 of 18 checks pass.

# 3 Part 1: what DMP can model or host, and the science it touches

## 3.1 The ledger

| Property of quantum gravity | Model | Host | Backing |
|---|---|---|---|
| Quantum content, correlations up to $2\sqrt{2}$ | no | yes | GUID check [5]; [B4] |
| Boundary entropy equals the minimal cut (Ryu–Takayanagi [24]) | on a tree | yes | GUID check [2] |
| The bulk as an error-correcting code [25, 26] | kinematics, one layer | yes | GUID checks [1]–[4] |
| Gravity entangles two masses [27, 28] | no | yes | GUID Claim A, check [6]; [B7] |
| Superposition of geometries (area sectors) | no | no, while records exist | GUID fact (v) |
| Classical limit: Lorentzian 3+1, Einstein's equations | open | yes | GUID X5 |
| Dynamics; $G$, $\hbar$, $\Lambda$ | no | as input | GUID fact (iii) |

So DMP's own structure is, at best, the classical part of a theory of gravity: the part that every reader agrees on. That places it, without any choice of ours, on the classical-gravity side of the current experimental programme. This is the only sense in which the verdict "can be used for real science": it commits DMP to predictions that experiments now in preparation can refute.

## 3.2 Consequence 1: no gravitationally mediated entanglement

Bose et al. [27] and Marletto and Vedral [28] proposed to put two masses, each in a superposition of two places, next to each other and let only gravity act between them. A quantum (coherent) gravitational field gives each branch the phase $Gm^2t/(\hbar r)$, with $r$ the distance in that branch; the combination of phases that no local operation removes entangles the masses. Galley, Giacomini and Selby [29] prove, in generalised probabilistic theories, that a field which entangles two systems by local mediation is not classical.

Check [B7] runs the geometry quoted by Bose et al. (masses $m = 10^{-14}\,\mathrm{kg}$, separation $\Delta x = 250\,\mu\mathrm{m}$ within each superposition, centres $d = 450\,\mu\mathrm{m}$ apart, closest approach $d - \Delta x = 200\,\mu\mathrm{m}$). The entangling phase grows at $0.126\,\mathrm{rad/s}$, so the negativity reaches $\sin(1/2)/2 \approx 0.24$ after $8.0\,\mathrm{s}$ and its maximum 0.5 after $25\,\mathrm{s}$. If the mediator is a DMP structure, that is, a record held in full by both halves, the branches are dephased and the negativity is 0 at all times.

**Proposition 1** (DMP's experimental commitment)**.** *In every dictionary in which a DMP structure is the gravitational field, gravity does not entangle two masses. A positive result of a Bose–Marletto–Vedral experiment refutes every such dictionary.*

Two caveats from GUID stand. Aziz and Howl [30] argue that classical gravity coupled to quantum matter fields can entangle, at a rate far below the quantum-gravity rate; this is disputed [31, 32, 33]. And the hosting half of the verdict is untouched by any such experiment: a configuration that hosts a quantum theory of gravity as content predicts whatever that theory predicts. The experiment tests DMP's structure, not DMP's content.

### 3.3 Consequence 2: if the classical geometry has dynamics, it must diffuse

DMP derives no dynamics. If one gives its classical geometry the only consistent dynamics we know for classical gravity coupled to quantum matter, Oppenheim's [34], a second test applies. That theory is linear in the density matrix, completely positive and trace preserving, and it has to be stochastic in both the metric and the matter. Oppenheim, Sparaciari, Šoda and Weller-Davies [35] show that it then obeys a trade-off: long coherence times of a mass in superposition require strong diffusion of the classical field, relative to the coupling. Interferometry on large molecules bounds the decoherence from one side; precise measurements of the acceleration of small masses bound the diffusion from the other, and together they already exclude part of the parameter space. The classical channel model of Kafri, Taylor and Milburn [36] makes a related, earlier prediction of a decoherence rate set by the gravitational coupling.

DMP adds no number to this. It picks the side: if its shared structures are the geometry and the geometry has any dynamics at all, it is in the class these experiments constrain. Torsion-balance measurements of gravity between masses below 100 mg [37] are the kind of experiment on the diffusion side.

### 3.4 Consequence 3: verdicts cannot certify randomness

This one is information theory and needs no new physics. Pironio et al. [22] showed that a Bell violation certifies fresh randomness without trusting the devices, and that nothing classical can. The Duality's verdicts are a local model ([B1], GUID check [5]).

**Proposition 2** (device independence lives in the content)**.** *No protocol whose outcomes are the Duality's verdicts can certify randomness or secrecy device-independently. A DMP configuration can host such a protocol only as content, with the gates reading that the content exists and never what it is.*

For an infrastructure built on DMP this is a design rule: device-independent randomness and key distribution belong inside a part, and the record of their use is counts.

## 4 Part 2: what the Bell experiments measured

Bell [2] showed that any theory in which each outcome depends only on its own setting and on a shared hidden state, distributed independently of the settings, obeys an inequality that quantum mechanics violates; Clauser, Horne, Shimony and Holt [3] gave it the testable form $|S| \leq 2$, where $S$ combines four correlations; quantum mechanics reaches $2\sqrt{2}$ [4]. A test closes a *loophole* when its design removes a way a local model could fake the violation: the *locality* loophole by space-like separating each setting choice from the other side's outcome, the *detection* loophole by recording enough events that no unfair sample can produce the violation, the *freedom-of-choice* loophole by making the settings as independent as possible of anything in the source (review: Brunner et al. [21]).

The 2022 Nobel Prize in Physics went to Aspect, Clauser and Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science". Their experiments and the loophole-free tests that followed:

| Year | Test | Outcome recorded on | Reported |
|---|---|---|---|
| 1972 | Freedman, Clauser [6] | photons (calcium cascade) | violation of the Bell-type bound |
| 1982 | Aspect, Dalibard, Roger [7] | photons; settings switched in flight | violation by 5 standard deviations |
| 1998 | Weihs et al. [8] (Zeilinger group) | photons; strict Einstein locality | violation |
| 2015 | Hensen et al. [9], 1.3 km | electron spins (NV centres) | $S = 2.42 \pm 0.20$, $p = 0.039$ |
| 2015 | Giustina et al. [10] | photons | $p \leq 3.74 \times 10^{-31}$ (11.5 s.d.) |
| 2015 | Shalm et al. [11] | photons | adjusted $p = 2.3 \times 10^{-7}$ |
| 2017 | Rosenfeld et al. [12], 398 m | rubidium atoms | $S = 2.22 \pm 0.033$, $p < 1.05 \times 10^{-9}$ |
| 2023 | Storz et al. [13], 30 m | superconducting qubits | $S = 2.0747 \pm 0.0033$, more than 22 s.d. |
| 2017 | Handsteiner et al. [14] | photons; settings from Milky Way starlight | settings fixed ∼600 years before |
| 2018 | Rauch et al. [15] | photons; settings from quasars | settings fixed at least 7.78 Gyr before |
| 2018 | BIG Bell Test [16] | 12 laboratories, several systems | settings from about 100 000 people |

The 2015–2023 tests close the locality and detection loopholes at once. Check [B6] records the three CHSH values: all lie above 2 and below $2\sqrt{2}$, at 2.1, 6.7 and 22.6 standard deviations above the local bound. The freedom-of-choice loophole cannot be closed, only pushed back: the cosmic tests require any local-realist influence on the settings to be older than the light from the quasars [15], and the BIG Bell Test replaces physical randomness by human choices [16].

## 5 Part 2: four readings of the Duality

A Bell test asks of a theory three things: what each outcome may depend on, whether the hidden state may depend on the settings, and whether the result can be used to signal. Each reading of the Duality below answers them differently. "Outcome" means what a laboratory records; Section 6 asks what that is under DMP.

**Reading (a): isolated halves.** Each half decides on its own setting and on what it read itself (Axiom 2), from a shared history that does not depend on the settings. This is `gateModel`, and `gate_obeys_chsh` proves $|S| \leq 2$. Check [B1] confirms it over all 16 deterministic strategies; every local model is a mixture of these (Fine [5]). *Refuted by the data* of [B6].

**Reading (b): full reach.** Axiom 1 gives each half "full reach over the act". If that means each half reads both settings, the gate reaches $S = 4$ (`rf_full_reach_reaches_four`; [B2] over all 256 strategies). Two facts follow. First, every deterministic strategy with $S > 2$ makes one side's outcome depend on the other side's setting ([B2]): it signals, and in the 2015–2023 tests the setting choices were space-like separated from the far outcomes, so the signal would be faster than light. Second, even a full-reach Duality that does not signal can reach 4: the Popescu–Rohrlich box [17] has uniform marginals and $S = 4$ ([B3]). Nature stops at $2\sqrt{2}$ ([B4]). To stop there a theory needs a further principle, such as information causality [18], and DMP has none. *Survives only with a principle DMP does not state, and then models nothing DMP adds.*

**Reading (c): a superdeterministic root.** Axiom 3 makes every layer a tree, and the Lean shows that the past of every line is a chain (`rf_past_is_a_chain`): all histories meet at the minting. A common past is what superdeterminism needs, that is, a hidden state whose

distribution depends on the settings (Hossenfelder and Palmer [20]). It is not enough: the root would have to correlate the hidden state with settings that, in the tests of Section 4, came from quasar light and from human choices. How much correlation is needed is a linear programme. With Hall's measure of measurement dependence [19], $M$, the largest total-variation distance between the hidden-state distributions of two setting pairs ($0 \le M \le 2$), check [B5] finds that the least $M$ with which a local deterministic model reaches CHSH value $S$ is exactly

$$M_{\min}(S) = \frac{S-2}{3},$$

that is, 0.025 for the ETH value, 0.073 for Munich, 0.14 for Delft, 0.276 for $2\sqrt{2}$ (about 14% of the maximum, matching Hall's figure for the singlet) and 2/3 for $S = 4$. Superdeterminism cannot be refuted by a Bell test in general, and we do not claim to refute it.

Inside DMP it has a second cost, which we argue but do not prove. Axiom 1 rests on two *independent* witnesses: witnesses that share no member of the declared failure classes. Measurement independence is the same kind of statement about a setting and the thing it measures. A root that correlates every later setting with the state it meets correlates every later pair of witnesses with the evidence they judge, and the minting is then a failure class that every pair shares. DMP can take this escape only by declaring that its own independence condition does not reach back to the root. *Survives logically, at a computed price, against DMP's independence.*

**Reading (d): hosting.** The Duality's verdicts are what DMP defines; the measured spins, atoms, qubits and photons are content inside parts, which the gates read only as existing. GUID's check [5] shows both at once: content at $2\sqrt{2}$, verdicts at 2. *Survives, and is GUID's verdict.*

## 6 Part 2: verdict on the hypothesis

**Hypothesis 1** (the first author, 29 October 2026)**.** "The 2022 nobel price winning bell experiment actually did not violate Bell inequality under Dualism."

**As stated, it does not survive.** The inequality was violated in the recorded data, by every loophole-free test, in four kinds of physical system ([B6]); a theory cannot make a measured $S = 2.0747 \pm 0.0033$ into a number at most 2.
**Sharpened, it does.** The form the evidence supports is:

**Claim 1** (the Duality never violated it)**.** *Under Dualism the outcomes the Bell experiments recorded are content, and content violates Bell's inequality up to $2\sqrt{2}$. The Duality's own verdicts, which DMP defines, are a local model and never violated it. The experiments therefore show that no DMP structure is the measured physics: what was measured is hosted.*

This turns a refuted hypothesis into evidence for GUID's verdict. GUID reached "cannot model, can host" from an argument about records and mediators; the Bell experiments reach the same edge from data, about a different property (non-locality instead of gravitational entanglement). Readings (b) and (c) are the only ways to make the Duality itself the measured physics, and each needs something DMP's text does not contain: a principle that stops at $2\sqrt{2}$, or a root that suspends its own independence condition.

## 7 The first author's premise on photons and gluons

The first author gave the following premise for this paper, in his words:

> "Photons and gluons are act carriers, so they are the only actual singular stuff, but because they don't have time, they don't exist. This means, instead of everything else that is ALWAYS BOTH, those are only ever a wave, except when they deliver their information, then they must exist and thus cease to."

*What matches.* A massless particle moves on a null worldline, along which no proper time elapses. A detected photon is absorbed: in every photonic Bell test the detection ends it. The photon and the gluon are exactly the gauge bosons of the two symmetries the Higgs field leaves unbroken, electromagnetism and colour, which is why both are massless [41]; this is the cleanest true form of "the only ones that are not always both".
*What does not.* (1) The premise cannot carry Hypothesis 1: three of the loophole-free tests recorded their outcomes on matter, electron spins [9], rubidium atoms [12] and superconducting circuits [13], which by the premise are "always both"; photons only linked the two sides. (2) Gluons are confined: they do not travel freely and are never detected alone, so "only ever a wave, except when they deliver" has no measured counterpart for them. (3) A photon is not "only ever a wave" in flight in any sense experiment can distinguish: its wave and particle aspects are both properties of one quantum, and which one a setup shows depends on the setup.

# 8 The first author's further hypotheses

## 8.1 The mass of the W and Z bosons

> "Since all bosons are IM packets, they all need to authenticate themselves. Photons do it by being quantized in the universal quantum. So since they are a multiple of something that exists, they must exist, and that's enough. Gluons are internal IM, because they are between quarks, and quarks are the first cluster that is actually being built. A quark is one node. Let's not go with part, actually, because part is singular in mathematics. Let's go with node, because one is at least always two, but can be more, like quarks, where mesons with two nodes are unstable, but three nodes, like in good Fedora cluster, is stable. And from there on, they build. And because gluons are internal IM, it's enough that that the information fits, and that they are internal, because internal must exist. And because if internal didn't exist, they wouldn't be, and because they are, it must. And since W and Z bosons are inter-cluster IM that must go through non-IM space, they need to carry both their information, and they must exist, so they need weight. So compared to all other bosons, W and Z bosons have weight, and because of that, and because they are duality, they also decay."

And, on the Higgs mechanism: "the Higgs mechanism, where something suddenly just exists because it needs to exist, just sounds like axiomatic correctness. So the bosons normally wouldn't have weight, but they need to have existence. So they have existence, then they deliver their information, and then they cease to have existence. That's all it is."
*The established result it touches.* The W and Z get their masses from the Brout–Englert–Higgs mechanism [38, 39] in the electroweak theory [41]: a field that fills all of space has a non-zero value in the vacuum, and the gauge bosons of the broken symmetry acquire mass by coupling to it. The same mechanism appears in a superconductor, where the photon acquires an effective mass inside the condensate (Anderson [40]). The Higgs boson was found at about 125 GeV in 2012 [42, 43]; the W mass is measured at $80.3692 \pm 0.0133$ GeV [44].
*What matches.* The shape: a carrier that must cross a medium filling space acquires mass. The first author's "non-IM space" takes the place of the Higgs field's vacuum value, and his reading of the mechanism as "axiomatic correctness" is a reading, not a disagreement.

*What does not, or not yet.* The mechanism predicts numbers the reading does not: the ratio $m_W/m_Z = \cos\theta_W$ at tree level, a scalar particle of its own, and couplings to that particle that grow with mass, all measured. "Three nodes are stable" holds for the proton (lifetime above $2.4 \times 10^{34}$ years in the channel $p \to e^+\pi^0$ [46]) but not for the free neutron (mean life $878.4 \pm 0.5\,\mathrm{s}$ [45]) or the other baryons, all of which decay; every meson decays, as the reading says. *Verdict: a re-description with the right shape and no testable difference yet.* It would get one if it fixed a number, the W/Z mass ratio first. The renaming of "part" to "node" is the first author's proposal for DMP's text; nothing in DMP changes without his approval.

### 8.2 Decay

> "The decay is actually just a principle thing, because the only things that aren't dualistic are photons and gluons, and as far as I know, they don't decay, and everything else has the ability to, as long as it becomes unstable. Since the electron is the, let's say, the chosen particle of the information field, I assume it's extremely stable, especially since it's kept stable through the fine structure constant."

*Test.* The photon and the gluon are not known to decay. "Everything else can decay once it becomes unstable" is true and close to a definition, so we file it as a principle, not a prediction. The electron is stable (lifetime above $6.6 \times 10^{28}$ years for $e \to \nu\gamma$ [47]), but in established physics for another reason: it is the lightest particle with electric charge, and charge is conserved, so nothing lighter can carry its charge away. The fine-structure constant sets how strongly it couples to light, not whether it lasts.

### 8.3 The fine-structure constant

The hypothesis rests on the first author's account of the start, the Oscillon Degradation Protocol (ODP), which he posted on DMP pull request #25 on 28 October 2026. We quote it as his cosmology; this paper does not test it, except where it meets the data below.

> "New Insight: That's why the ODP is needed! G0 only has information on how to Exist, so it's first act must be to find out how to act! That's why the perfect duality MUST decay, since it only axiomatically can interact with itself, since it bounds what Exists. This produces a true naked duality, since it would normally be bounded by an EH, since it can only interact with itself, but that boundary is already spanned by it having to exist, since it does. It thus radiates out HPE at 100% efficiency, which inflates the Existence. The Oscillons, which then bound Existence, but are themselves only Existence, must X to Existence, but have nothing TO X to but Existence, so they output both the Information required to act and the Existence required to act on it, which, through reheating, leads to today :)"

His correction of 29 October: "HPE was wrong, it is just Information and Existence."
On the fine-structure constant itself, in his words:

> "It started as one because at the beginning, existence and information had to be bound maximally together to force an act. And from then, from the first two acts, so the first act that wasn't forced because something else already exists to interact with, from there the fine structure constant decreased until today, where it's pretty stable because existence and the information it needs to assert itself are in pretty much in balance. But it will get slightly lower still because the inflation is slowing down, so more assertion per unit of space-time is possible."

*Test.* This is a prediction, and most of it is measured. At recombination, about 380 000 years after the big bang, the cosmic microwave background gives $\alpha/\alpha_0 = 0.9934 \pm 0.0042$ [48]. Quasar absorption spectra agree with today's value to about a part per million over redshifts 0.6 to 2.4, roughly the last ten billion years (combined $-0.5 \pm 0.5$ ppm, ESPRESSO [49]). So a fall from 1 to 1/137 would have to happen almost entirely before recombination.
The first author then placed the fall there, in these words (the bracket is our reading of a dictated word, after the project's name for its oscillon work):

> "Since by the time of the hot Big Bang and the cosmic microwave background, existence and information were already existent enough to assert themselves with each other, most of the fall from one to today's number will probably happen would have happened before that, so in the reheating phase, because that is what actually forced the reheating, because after the Ocelons [oscillons] decayed, there was only existence and information, but not combined, so they had to combine, because that was the only thing they could do, and through that, the first dualistic existence by itself was formed, and as soon as another was formed it could interact with, it was no longer a fine structure constant of one, but slightly less than one, and from there it decreased until it always is kept stable, because it has to, because an act must happen, because it exists, so it adjusts to what it needs to."

The earliest probe of $\alpha$ is big-bang nucleosynthesis, seconds to minutes after the big bang; recent analyses bound $\Delta\alpha/\alpha$ there at the level of tens of parts per million, the number depending on the unification scenario assumed [50]. Reheating ends before nucleosynthesis begins, and oscillons are a studied feature of the end of inflation [51]. So a fall during reheating is consistent with every measurement we found, at the price that no measurement of $\alpha$ in the past can test it. The prediction that $\alpha$ "will get slightly lower still" is live: optical clocks measure today's drift as $(1.0 \pm 1.1) \times 10^{-18}$ per year [52], consistent with zero, and the hypothesis predicts a negative value below that limit. *Verdict: a decrease after nucleosynthesis is excluded; a decrease during reheating is consistent with the data and untested by them; the future decrease is a testable sign prediction.* "It adjusts to what it needs to" is a stabilising clause and predicts no number by itself. One established fact has the hypothesis's shape: the effective coupling of electromagnetism grows with energy (it is close to 1/128 at the Z mass), so the hot early universe saw a larger effective $\alpha$; in physics that runs with energy, not time, and it does not reach 1.

## 9 Where we expect it to break

1. *Reading (b).* Find a sentence of DMP's text that bounds full reach at $2\sqrt{2}$ without signalling (for instance by information causality [18]). That would make the Duality itself a candidate model of the measured correlations.
2. *Reading (c).* Show that DMP's independence of witnesses is compatible with a root that correlates every setting with its hidden state, or that the root needs less than $M_{\min}(S)$ because DMP's hidden states are not a mixture of deterministic strategies.
3. *The linear programme.* [B5] uses CHSH with two settings per side and Hall's distance; another measure of measurement dependence gives another number. The equality $M_{\min} = (S-2)/3$ is checked at six values, not proven.
4. *Proposition 1.* It falls with GUID's Claim A and weakens if classical gravity can entangle [30].
5. *Section 8.* Each verdict there rests on the measured value we cite; a better measurement of the drift of $\alpha$ with a negative sign would turn the last one into support.

## 10 Conclusion

What GUID's verdict buys is a commitment: DMP's own structure is classical, so if it is the geometry, gravity does not entangle masses, and if that geometry has dynamics, it diffuses; experiments now in preparation can refute both. The Bell experiments of the 2022 Nobel Prize did violate Bell's inequality, and under Dualism that is what one should expect of content. The Duality's verdicts never violated it; the measured particles did. Two very different experiments, one about gravity and one about non-locality, point to the same edge: *host, not model.* The first author put the stance in one line: "Axiomatically, since we already don't have all the information that was contained at the start, we can only know existence and can know no more about our reality. So axiomatically, we can infer all that we can and no more."

## Contributions and use of AI

Steven Hrovat conceived DMP, set the questions of this paper and gave the hypotheses quoted in Sections 6–8 in his own words; he takes responsibility for the axiom system. Sirius (Claude Opus 5.5, Anthropic) checked each hypothesis against the literature, wrote and ran `docs/bell/checks.py`, and wrote this text. Every reference was checked against a publisher, arXiv or collaboration record on 29 October 2026; the record is `docs/bell/README.md`.

## Licence and availability

This paper and its source are released under the Apache License, Version 2.0. DMP: `https://github.com/StevenHrovat/dmp` (version 0.9.9 at commit `4400cce`). Gravity: `https://github.com/StevenHrovat/gravity` (this paper in `docs/bell/`; GUID in `docs/guid/`; the Lean development in `evidence/`; `gate_obeys_chsh` on pull request #17).

## References

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